

EXPERIMENTAL AERODYNAMICS

PROJECT PORTFOLIO

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Wind-Tunnel Testing • GLOF • TSP • Optical Flow • Skin-Friction & Surface
Pressure • Force Measurement



Experimental Aerodynamics & Flow Diagnostics

Focused on designing and conducting aerodynamic experiments, extracting quantitative flow measurements, and validating results through independent measurement techniques.

WIND TUNNEL

test planning • setup • execution

GLOBAL DIAGNOSTICS

GLOF • TSP • OFV • OFI

MEASUREMENT

pressure • load cells • DAQ

DATA REDUCTION

MATLAB • calibration • validation

What I bring to an applied aerodynamics team

Develop and refine experimental facilities and measurement systems.
Turn optical images into skin-friction, pressure, heat-transfer, and flow-field quantities.
Design calibration and validation paths instead of treating diagnostics as black boxes.
Communicate results through peer-reviewed publications and cross-functional test work.

4.0

Ph.D. CGPA

100+

students taught

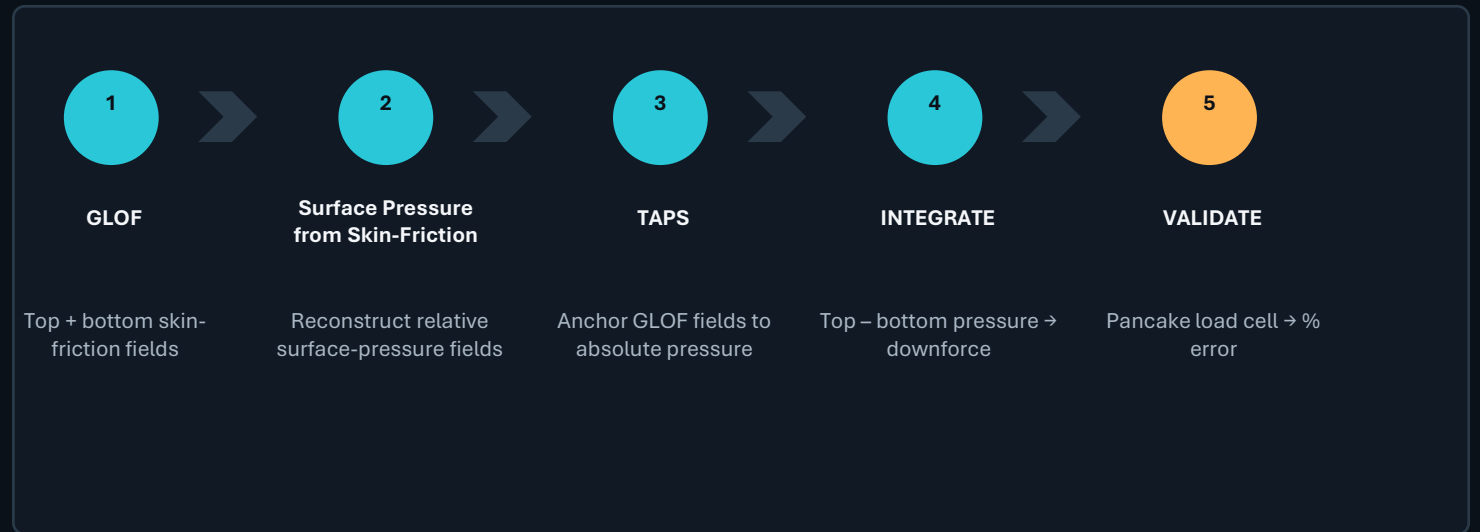
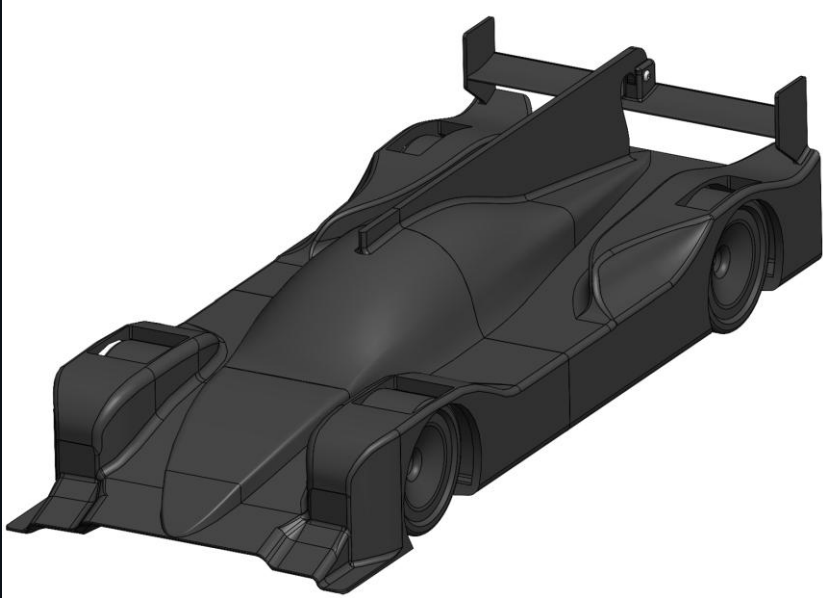
4

selected
publications

Experimental aerodynamics • wind-tunnel testing •
image-based flow diagnostics

Le Mans vehicle: Global surface aerodynamics to downforce(on-going)

A single experiment chain connects GLOF fields, pressure taps, integrated pressure load, and force-cell validation.

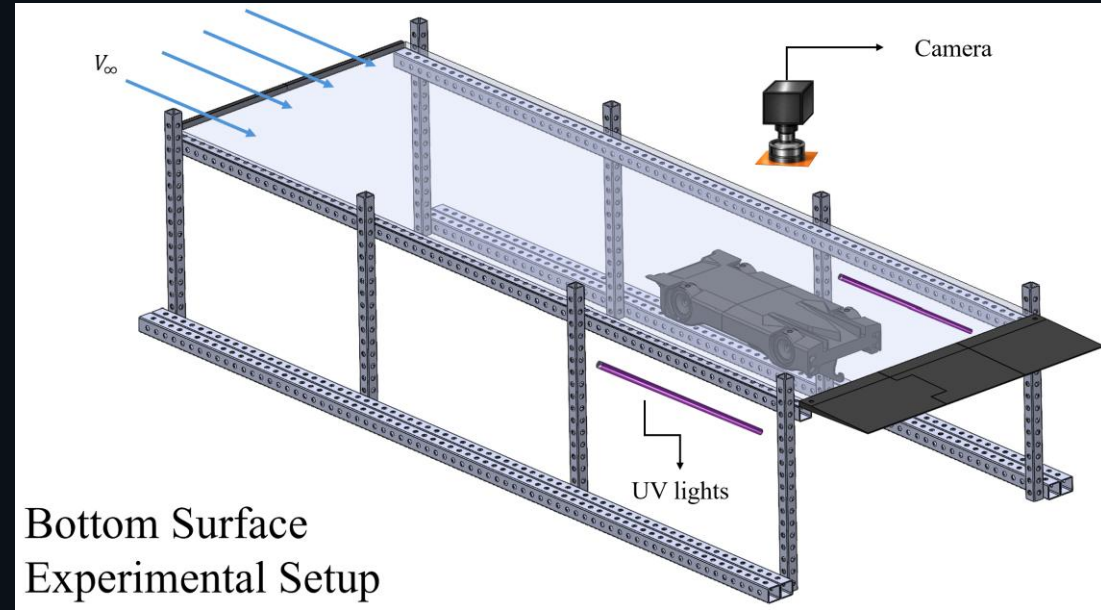
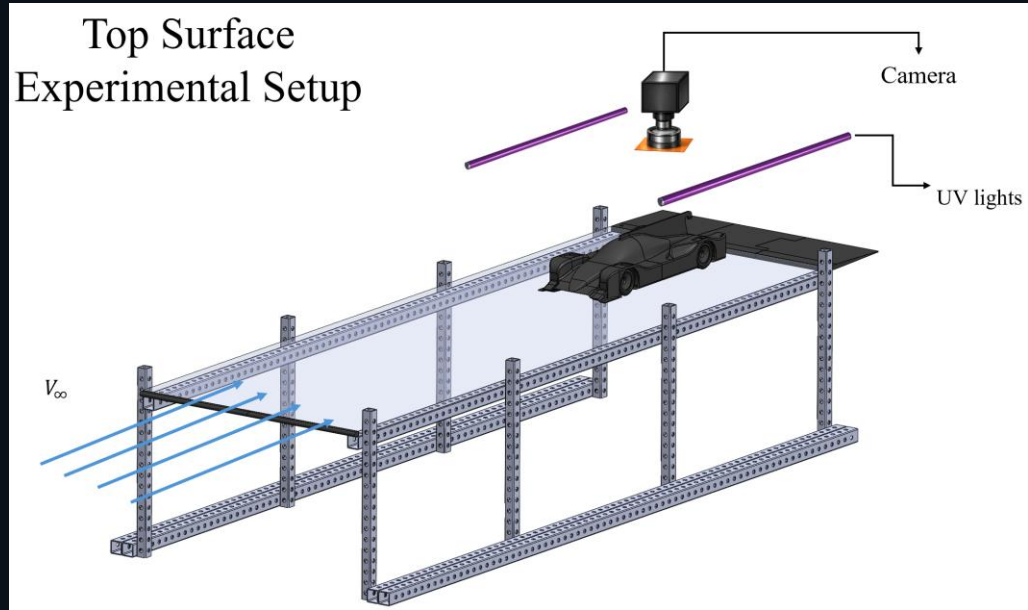


WHY THIS MATTERS The project is not only a visualization exercise. It creates a traceable route from image-derived surface physics to an independently measured aerodynamic load.

Status: GLOF top/bottom acquisition and processing underway; pressure-tap anchoring and pancake load-cell validation are the next validation layers.

Test configuration designed for top and underbody access (on-going)

The bench preserves the ground-reference geometry while allowing optical access to both surfaces.



TOP-SURFACE GLOF

Camera above model + UV illumination

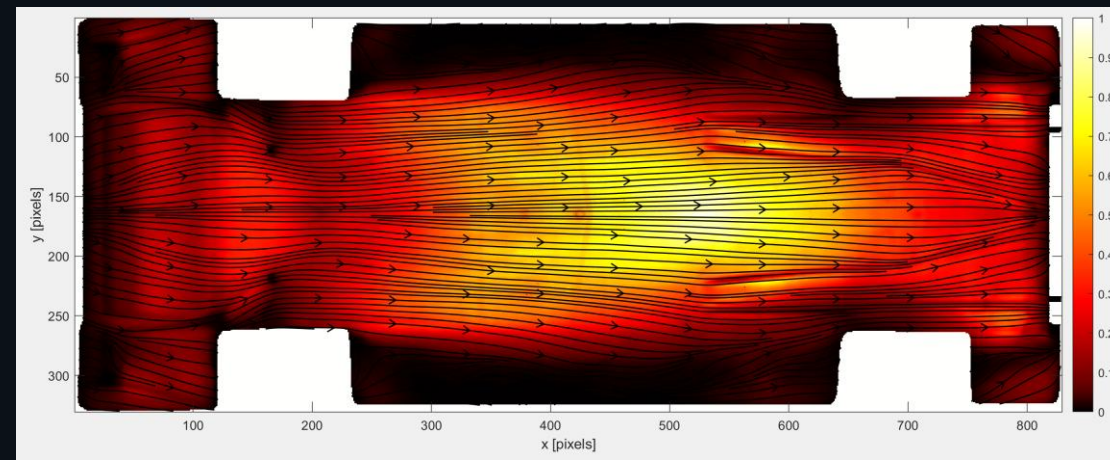
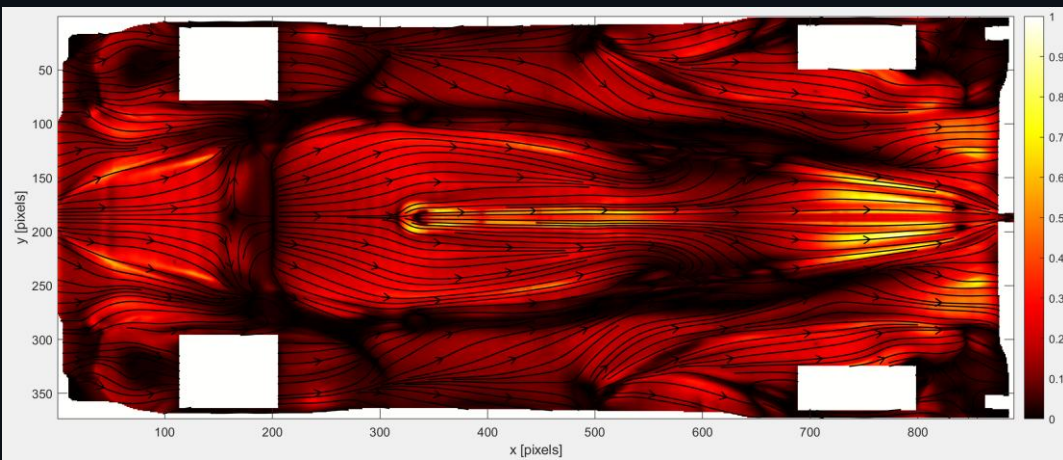
UNDERBODY GLOF

Optical access below the acrylic ground plane

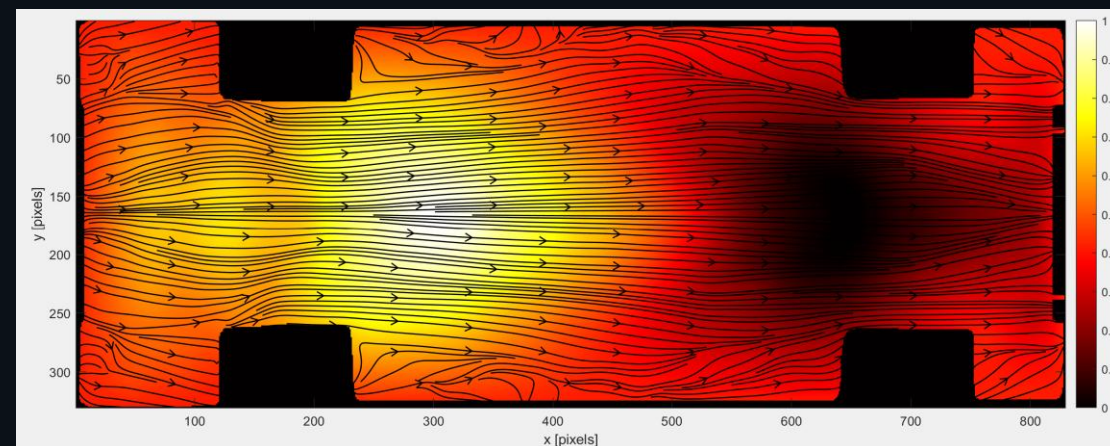
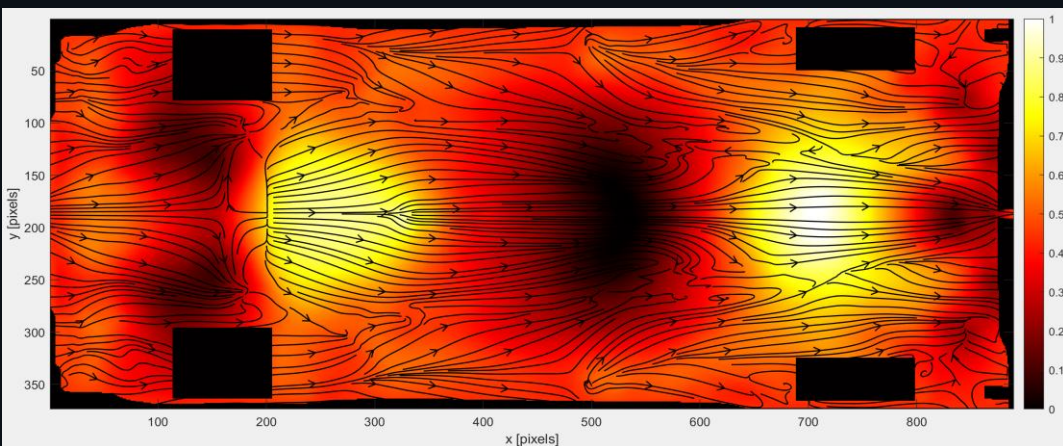
Upper Surface

Skin - Friction Field

Bottom Surface



Surface Pressure Field



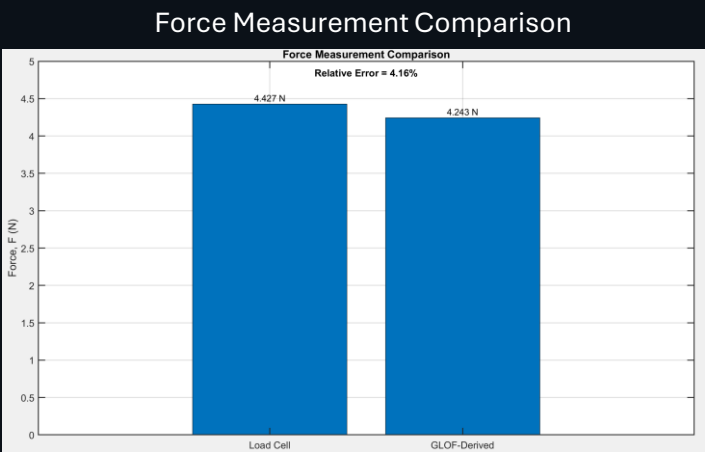
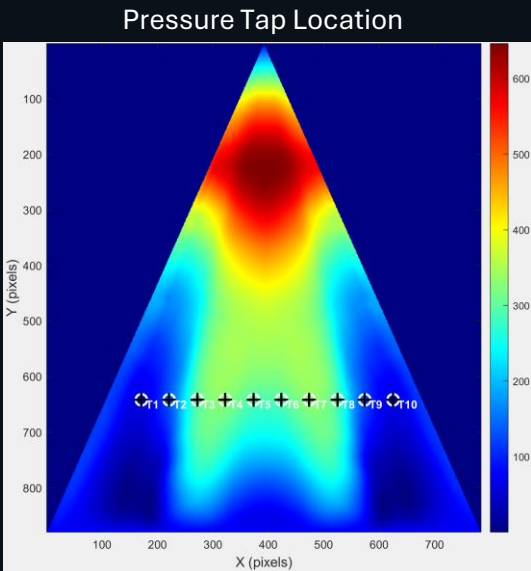
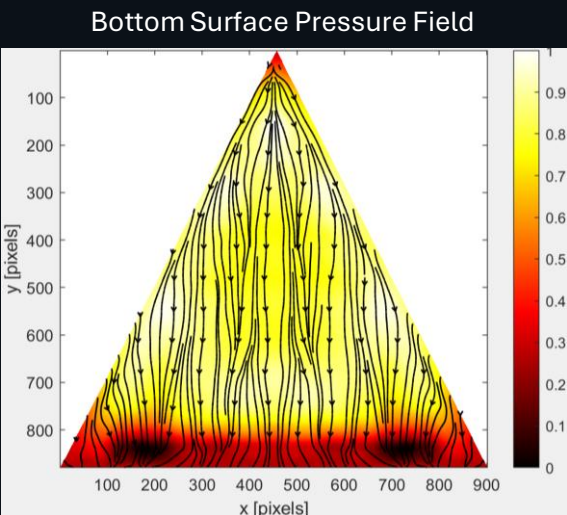
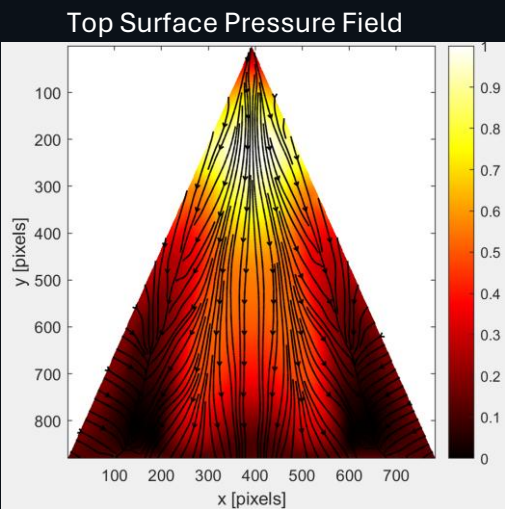
Delta wing: pressure integration validated against force measurement (on-going)

The same GLOF → pressure → integrated-load logic has already been demonstrated on a simpler aerodynamic geometry.

4%

error between integrated pressure-derived load and independent force measurement

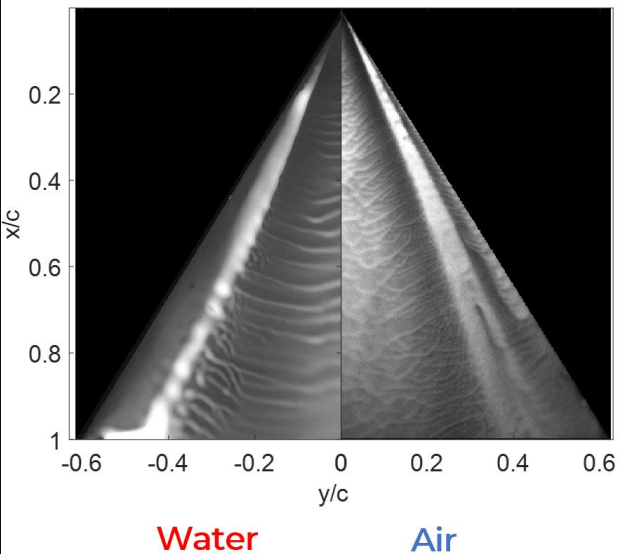
GLOF used to reconstruct surface quantities.
Discrete pressure taps provide absolute pressure anchors.
Pressure field integrated over the wing surface.
Independent load measurement provides a direct validation target.



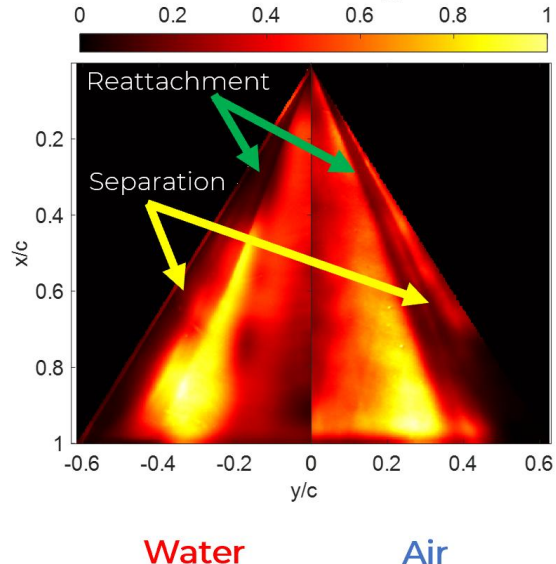
GLOF in water: same near-wall physics, different medium

Stress test of the diagnostic: high-resolution skin-friction topology was reconstructed in water and compared with wind-tunnel measurements.

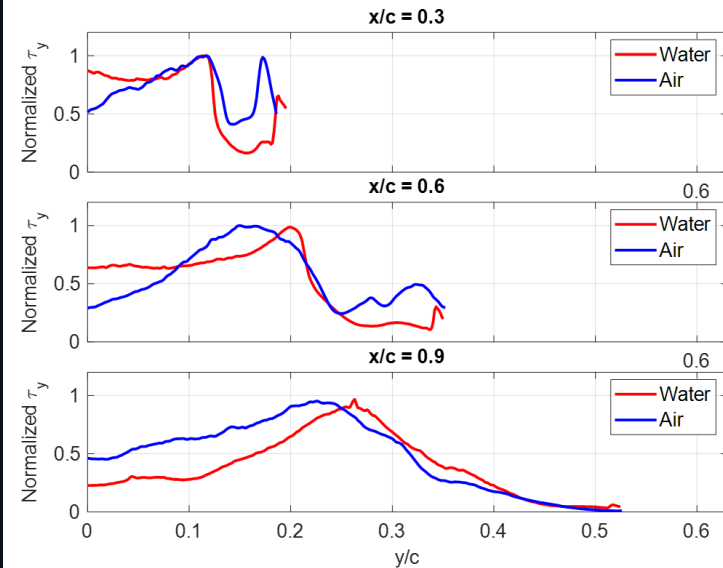
Sample GLOF Images



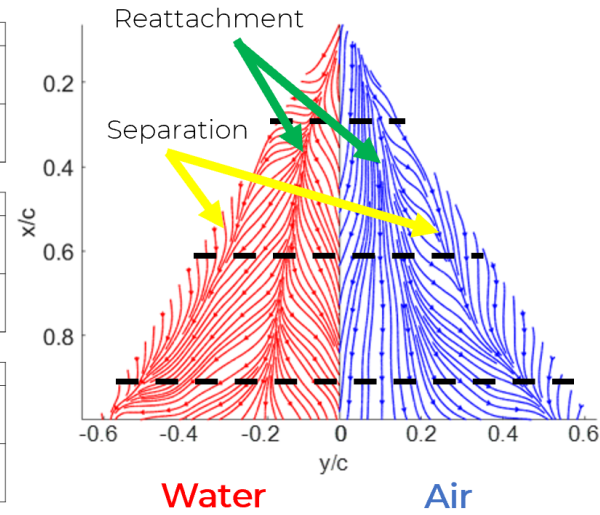
Skin Friction Magnitude



Skin Friction Profiles



Skin Friction Lines



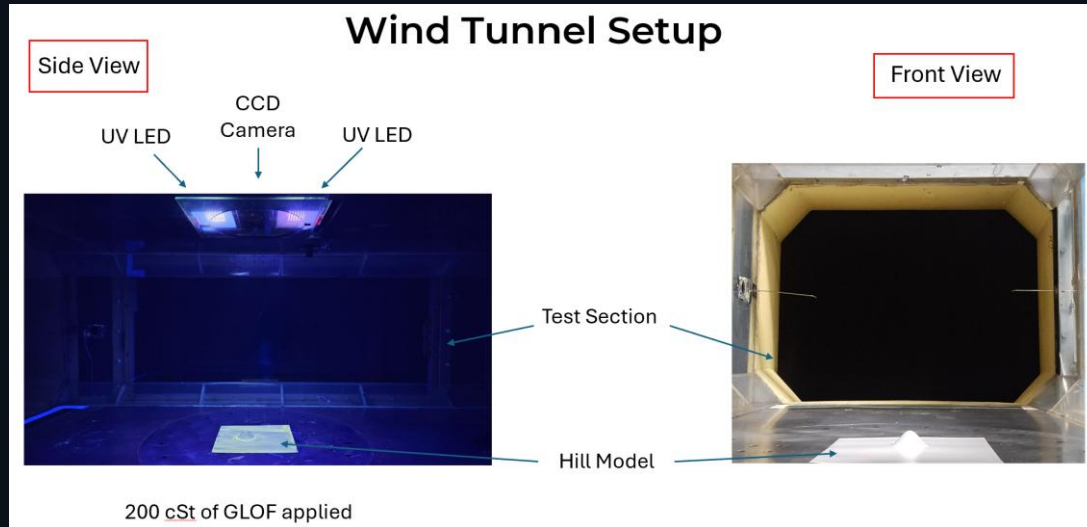
RESULT

Separation and reattachment structure was recovered from the skin-friction field, and the delta-wing result was quantitatively consistent with the air experiment.

Why it matters: it shows the diagnostic is a transferable measurement framework, not a one-off visualization trick.

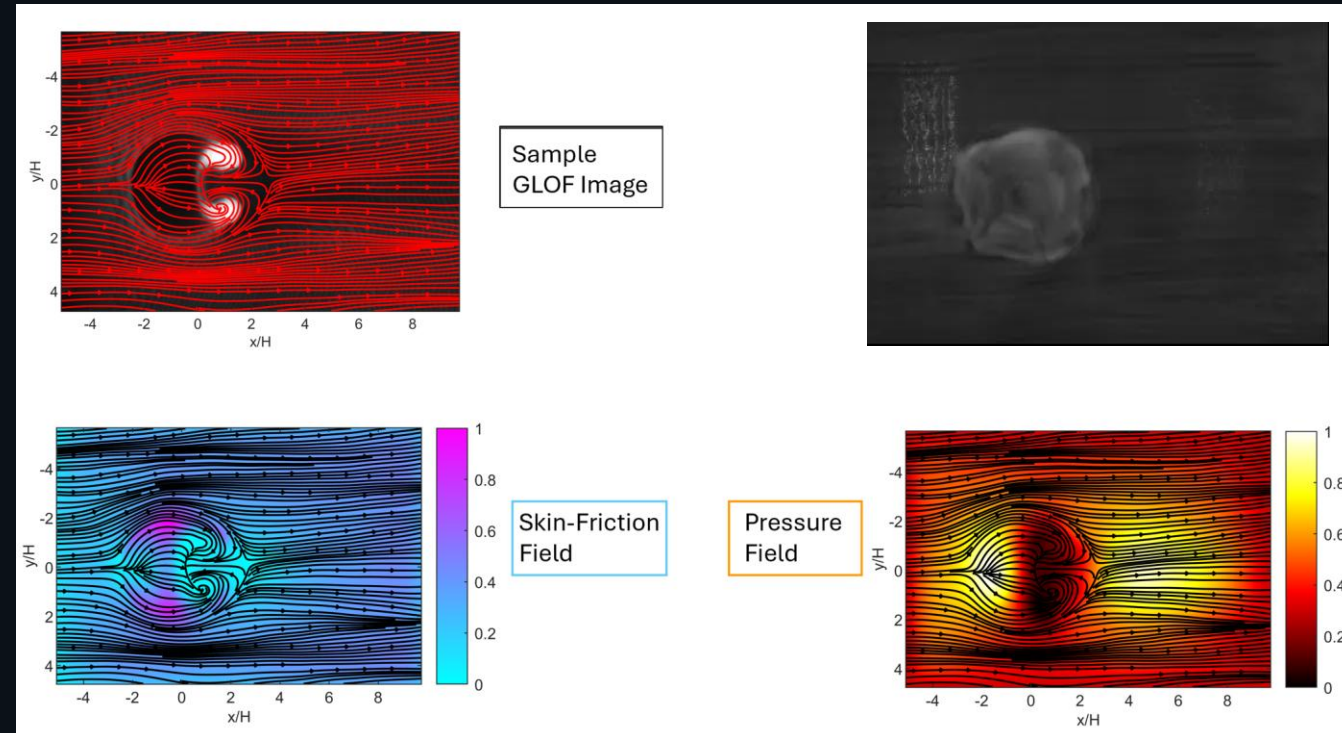
NASA FAITH hill: skin-friction topology + pressure reconstruction

Global surface measurements reveal the near-wall structure around a canonical three-dimensional hill.



MEASURED / DERIVED

GLOF skin-friction field
 Surface-pressure field derived from skin friction
 Nodes, saddles, separation and reattachment topology
 Reynolds-number and geometry trends



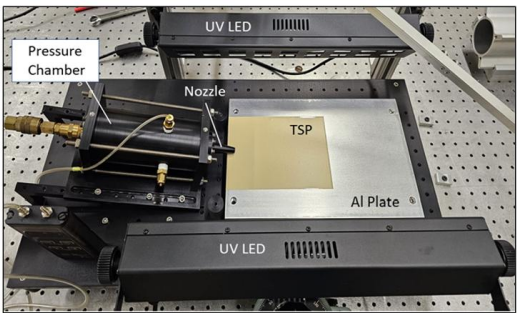
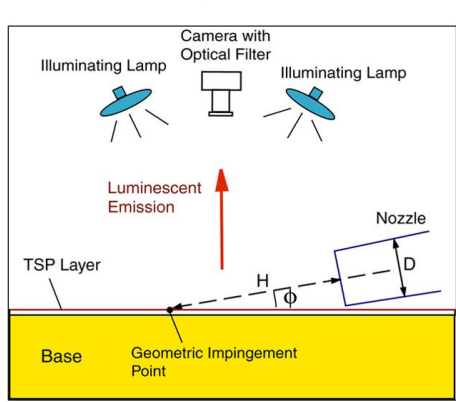
Physics of Fluids, 2026

Near-wall structures of topographic hill flow: skin friction, surface pressure, and derived quantities

Oblique impinging jet: TSP from temperature to heat flux and skin friction

A Mach-number sweep from 0.1 to 1.0 combined luminescent thermometry, image processing, and derived near-wall quantities.

Schematic of Experimental Setup

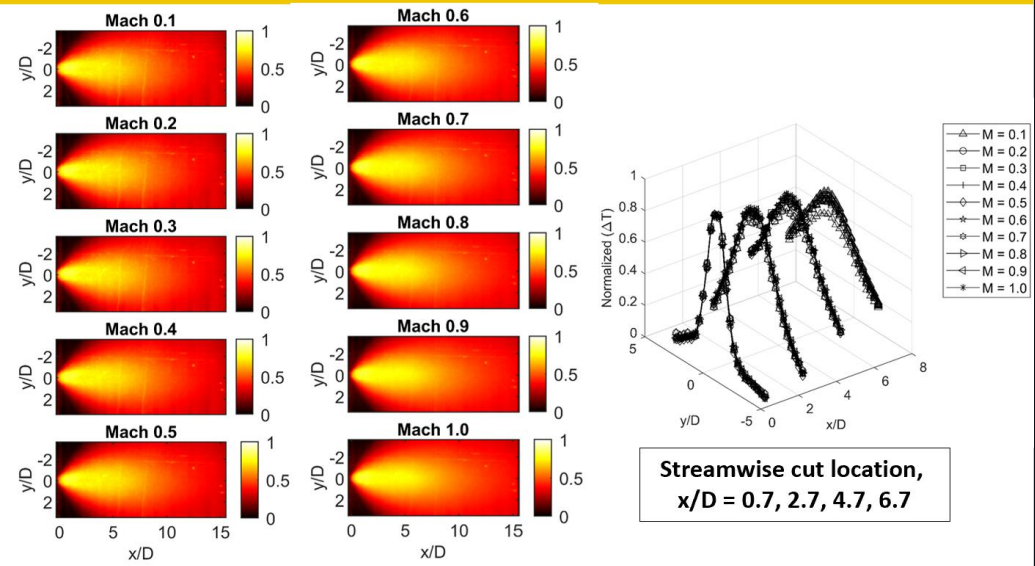


Experimental Setup

PIPELINE

TSP intensity → surface temperature → heat flux →
skin-friction topology → surface pressure
extraction from skin-friction → RMS fields

Surface Temperature from TSP ($H/D=6$)



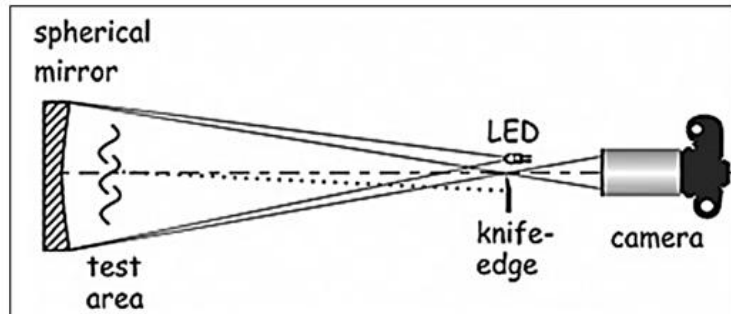
Selected result: normalized temperature and heat-flux profiles collapse across operating conditions, supporting self-similarity.

PEER-REVIEWED

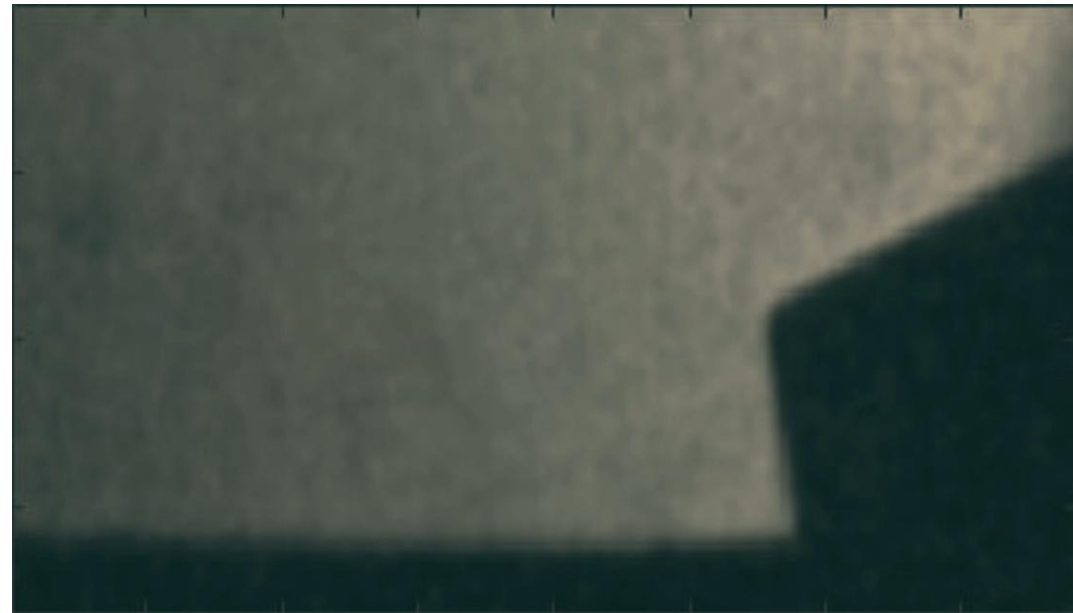
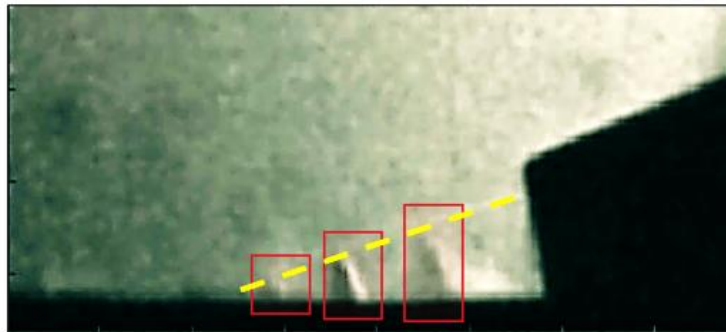
International Journal of Heat and Mass Transfer, 2024

Oblique impinging jet: Schlieren Imaging Capturing Shockwave

Experimental Setup



Shockwave Captured at $M=1$



Cummins: instrumentation, correlation, calibration, execution

TEST PLANNING

Developed test plans and executed procedures to validate system functionality.

INSTRUMENT CORRELATION

Evaluated QCL / PEMS / particle-number systems against laboratory references.

CALIBRATION

Created standardized calibration and experimental-procedure guides for technicians.

CROSS-FUNCTIONAL

Worked with engineers and technicians in RTE and CVS test-cell environments.

Planned Experimental Studies

Upcoming parametric studies extending global surface measurements to rotation-driven and separated flows.

ROTATING JUNCTION FLOW

Spin-ratio effects on near-wall aerodynamics

- Investigate the effect of spin ratio on skin-friction and surface-pressure fields.
- Quantify rotation-induced momentum transfer and changes in junction-flow topology.
- Primary control parameter: spin ratio.
- Status: experimental study in development.

BACKWARD-FACING STEP

Step-height effects on separated flow

- Investigate the effect of step height on skin-friction and surface-pressure fields.
- Quantify separation-bubble size and downstream reattachment location.
- Compare near-wall topology across multiple step-height configurations.
- Status: experimental study in development.

Methods I can bring into a fast experimental program

A mix of optical diagnostics, conventional instrumentation, data reduction, and aerodynamic interpretation.

EXPERIMENTAL

Wind-tunnel setup & test execution
GLOF skin-friction / pressure diagnostics
Temperature-sensitive paint (TSP)
Pressure taps & differential pressure
Load cells / aerodynamic force measurement
High-speed scientific imaging
DAQ, calibration, uncertainty & validation

COMPUTATIONAL / DATA

MATLAB image processing & field reconstruction
Physics-based optical-flow / variational methods
Surface integration and force estimation
Visualization, calibration fits, error analysis

SELECTED PUBLICATIONS

International Journal of Heat and Mass Transfer • Physics of Fluids
• SAE International • AIAA Aviation Forum

Research spans heat transfer, near-wall surface physics, optical diagnostics, and instrument validation.

Why this portfolio fits applied aerodynamics

The through-line is practical experimental engineering: design the test, make the measurement trustworthy, and turn the result into a decision.

Vehicle / aerodynamic analysis

Le Mans downforce workflow + delta-wing load validation

Low- and high-fidelity modeling

Physics-based global diagnostics + quantitative data reduction

Wind-tunnel experiments

Facility setup, optical access, calibration, troubleshooting, execution

MATLAB & data analysis

Image processing, field reconstruction, integration, uncertainty

Cross-functional engineering

Research teams, teaching labs, Cummins test-cell programs